

1) What is Power BI and how does it differ from Excel?

Power BI is a visualization tool where we can change the complex data into the attractive dashboard.

We use power BI for data analytics, business analytics.

Power BI is also called business intelligence tool.

In the power BI we can clean the data by using power query.

We can import data in power BI from many servers like Excel workbook, CSV, SQL, Azure, Python etc..

Power BI is different from Excel as we can load a large data set and easily work on it.

When we load a large data set in Excel, it may not work properly, it slows down, there is also a chance of data interruption.

When we use power BI on a data we connect with the server, so on the same data many people can work on it,

If many people have to work on the same data in Excel we have to share the Excel file.

Explain the concept of data modeling in Power BI.

Suppose we have a data set in that data we have five tables and we have to analyze all five tables together we need to establish a relation among them. At that time we use data modeling.

Power BI automatically builds some relation among the tables.

If we want to build a new relation we use data modeling, but there is one thing both tables must have a common column; otherwise we can create a new relation.

What are the different types of connections available in Power BI?

In the power BI there are three types of connections.

1) Import mode 2) Direct query mode 3) Live connection

Import mode: In the import mode the data is stored in the power BI.

In the import we can store small data sets.

We can load data up to 2GB.

Direct Query Mode: In this mode data is stored on the server.

We can store the data more than 2gb

Live connection: in this mode we can connect over data to live server, like sql

Our data is automatically updated when it's updated on the server we need to just refresh our data

How do you handle data transformation in Power BI?

When I get data first of all I scrutinize the data

I check the column quality of all the data it helps me to find where is error, null value, blank value,

After this I go for the cleaning the data and then as per requirement I built some kpi, chart, table etc....

What is DAX (Data Analysis Expressions) and why is it important in Power BI?

In the Power BI DAX is DATA ANALYTIC EXPRESSION

The DAX is used to create new column, new measure.

DAX helps us to perform advanced calculations

DAX helps us to create a dynamic report

6) Can you explain the difference between calculated columns and measures in Power BI?

When we use measure it creates a particular value which will appear in our report view.

But when we use new column it will create a new column in our table view. It also appears in the report view.

7) How do you handle relationships between tables in Power BI?

As we know that Power BI built automatically relations between columns

If we have to create a new relation then go to the model view and find common columns between tables and create a new relation

8) What is the purpose of a Power BI Gateway?

The purpose of a Power BI Gateway is to act as a bridge between on-premises data sources (like SQL Server, Oracle, SAP, etc.) and

the PowerBI Service in the cloud. It enables secure data transfers so users can refresh their reports and dashboards without manually uploading data. Essentially, it keeps cloud-based reports up to date with on-premises data.

9) How can you schedule data refresh in Power BI Service?

*To schedule a data refresh in Power BI Service, you go to the **workspace** → **Datasets** → **Settings** → **Scheduled refresh**. From there, you can enable scheduled refresh, set the frequency (daily, weekly, multiple times a day), and specify the time zone. If the data source is on-premises, a Power BI Gateway must be configured to allow refresh.*

10) Explain the concept of row-level security in Power BI.

Row level security is a one kind of security

Suppose we have a report and we have to share the report with different people, but we want some people don't have authority to access the whole report that means we want to restrict them in this case we use row level security.

11) What is the Power BI Desktop and how does it differ from Power BI service?

Power BI Desktop is offline mode we can store our data report in our desktop

Also in the Power BI Desktop we have less number of visualization charts as compared to Power BI service

Power BI service is online mode where our data is linked to the server

In this mode our data is stored on the server

That allows us to use the data many people

We don't have to share the data when we use Power BI service

We just have to give access to the person

12) Explain the concept of Direct Query in Power BI.

When we use direct query mode we can use a large data set without any interruption

Its allow to store our data on the server

Its also allow to import data from the server

13)What are Power BI templates and how are they useful?

Power BI templates are .PBIT files that store the structure of a Power BI report including queries, data model, relationships, measures, and visualizations but exclude the actual data. They are useful for reusability and consistency because you can share a template across teams, and different users can connect it to their own data sources while maintaining the same report design and logic.

14) How do you handle incremental data refresh in Power BI?

Incremental data refresh in Power BI allows you to update only the recent data while keeping the historical data unchanged. This is configured using parameters (RangeStart and RangeEnd)in Power Query, and then enablingIncremental Refresh Policyin the dataset settings. It improves performance, reduces refresh time, and is especially useful for large datasets.

15) What is the role of Power Query in Power BI?

Power query is allow us to clean our data , mange our data

Suppose we have to create a column then we use power query in that query we use custom column

Power query allow us to mainuplate the data according to our need

16)Explain the difference between calculated columns andcalculated tables in Power BI.

When we use create column it create a single column

When we create a table it create a new table where multiple columns and roe are presnt

17) How do you create custom visuals in Power BI?

When we have to create custom visual we have an option that is called report view. In that option we have multiple charts that allow us to show the characteristics of the data.

Suppose we have sales data of a year and we want to find sales trend. We use a line chart that gives us a visualization of sales trend.

18) What are the best practices for optimizing performance in Power BI? To optimize performance in Power BI, we should focus on optimizing the data model, queries, DAX calculations, and report design. Some best practices include using star schema, reducing data volume, avoiding complex DAX, and using aggregation tables. Also, leveraging features like incremental refresh and proper use of relationships improves performance."

19) How can you integrate Power BI with other Microsoft products like Azure and Office 365?

Power BI integrates seamlessly with other Microsoft products like Azure and Office 365 to provide end-to-end business intelligence solutions. It connects to Azure for data storage, analytics, and AI, while Office 365 integration allows collaboration and sharing through Teams, Excel, and SharePoint."

Azure SQL Database / Azure Data Lake / Synapse → as data sources for Power BI.

Azure Analysis Services → Live connection for enterprise data models.

Azure Machine Learning → integrating predictive analytics and AI models into Power BI.

Azure Active Directory (AAD) → manages security, authentication, and row-level security.

Excel → Import data models, use "Analyze in Excel" from Power BI.

SharePoint Online → Embed Power BI reports on SharePoint pages.

Teams → Share and collaborate on Power BI dashboards directly inside Microsoft Teams.

Outlook→ Share reports and snapshots via email; embed Power BI visuals in Outlook.

Power Automate(part of Office 365 ecosystem) → Automate workflows like triggering alerts or emails based on Power BI data.

20) Explain the concept of aggregations in Power BI.

Aggregation in power bi is mathematical operation

When we have to find sum, product, division, average, min, max, mode, etc.... these terms are known as aggregations

21) How do you handle error handling and data quality in Powerbi?

When I have data in that data when we check column quality if it shows that there is error in some columns

1st I will identify that which type of the data has the particular column

Suppose if a column has a data of average in that column there are some error

Now I want to remove that error I will go with fill up and fill down

Suppose I have data of selling price and that column some error exists in that case I will replace that error by not available

Similarly if I have data of selling product and in that column there are some error then I will replace that value by not available

22) What is the purpose of Power BI Embedded and when would you use it?

Power BI Embedded is a Microsoft Azure service that allows developers to embed Power BI reports, dashboards, and visuals directly into custom applications, websites, or portals. It provides business intelligence capabilities to end users without requiring them to use the Power BI Service separately."

To deliver analytics within external applications(customer-facing apps, vendor portals, partner systems).

To provide seamless user experience—users don't need a Power BI account.

To allow ISVs (Independent Software Vendors) and developers to integrate data visualization directly into their software products.

To ensure scalability and cost-effectiveness—pay for usage through Azure capacity.

When building a customer portal where customers should see reports but don't have Power BI licenses.

When a company wants to embed analytics in their own applications for clients (e.g., an ERP, CRM, or e-commerce platform).

When you want control over the look and feel of Power BI visuals inside an app.

For multi-tenant applications where each client needs to see their own data securely.